Unsupervised Learning and K-Means Clustering

GENERAL ASSEMBLY

Supervised vs. Unsupervised Learning

Almost all machine learning models can be divided into two camps:

- **Supervised learning models** - Models in which we use *x*-variables to predict some *y*-variable. For example...

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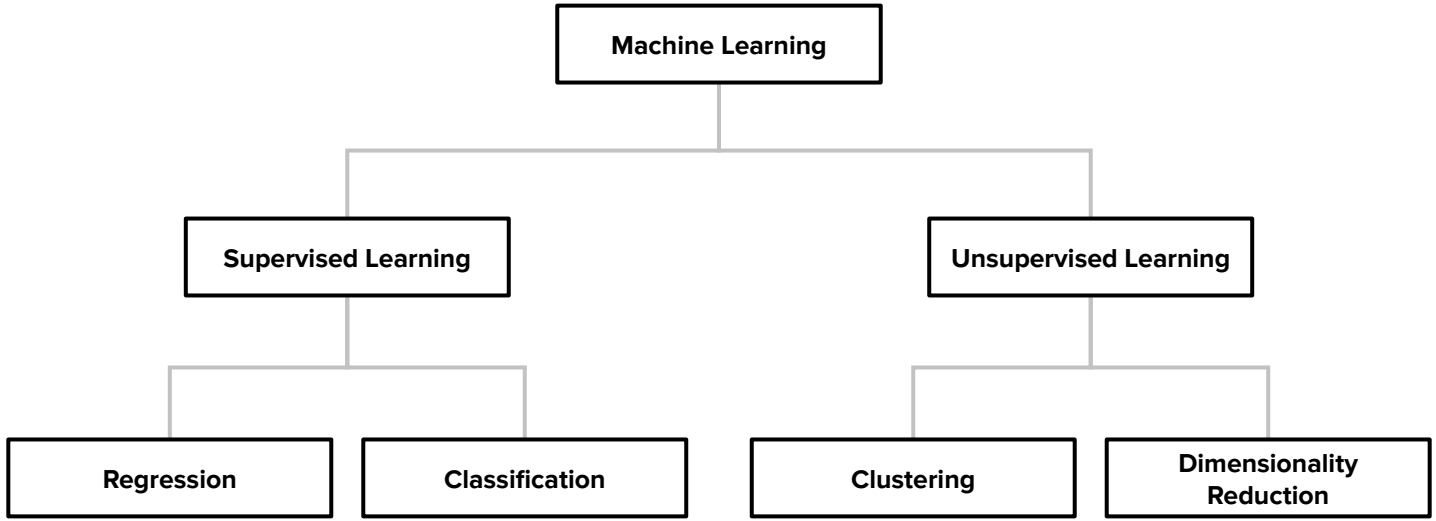
Supervised vs. Unsupervised Learning

Almost all machine learning models can be divided into two camps:

- **Unsupervised learning models** - Models in which we only have *x*-variables, and we are not trying to predict something. Instead, we are exploring the data we have to discover new relationships. For example...

"What do our customers who shop on Sunday look like? Are they different from our customers who shop online?"

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Machine Learning
Supervised Learning
Unsupervised Learning
Regression
Classification
Clustering
Dimensionality Reduction

What is Clustering?

Clustering is a task in which we seek to group our observations into **homogeneous groups**, or **clusters**.

Remember: since this is unsupervised, we don't decide in advance what these clusters will look like. It's all up to the algorithm!

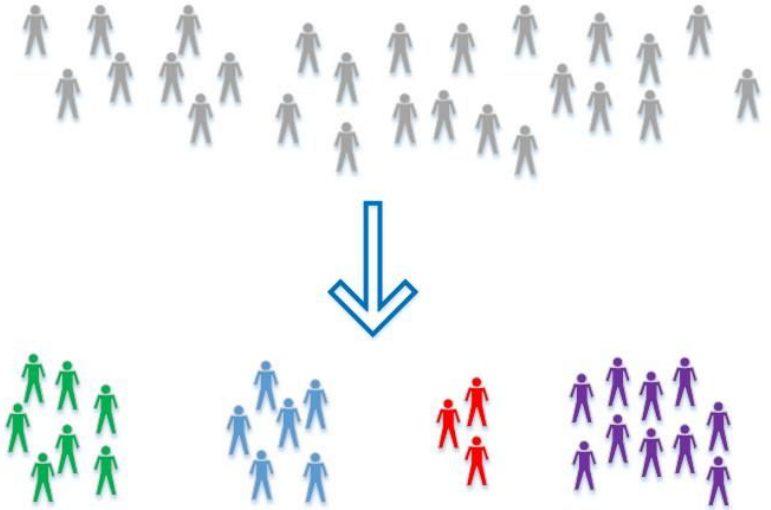
What is clustering used for?

There are several reasons why you might choose to group your data into similar bins. For example...

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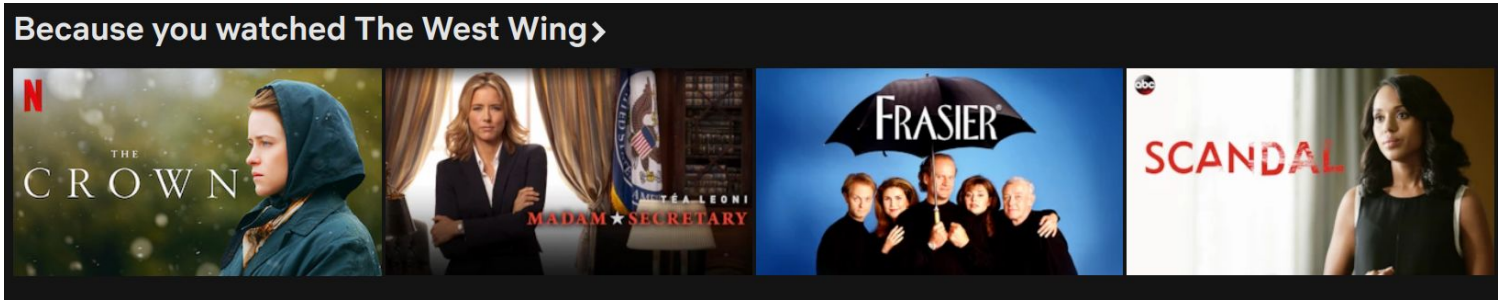
Market Segmentation

Clustering your user base is a popular use case for clustering. Suppose after clustering, you realize people in one cluster aren't buying your product as often as other clusters. Why? What does this cluster look like?

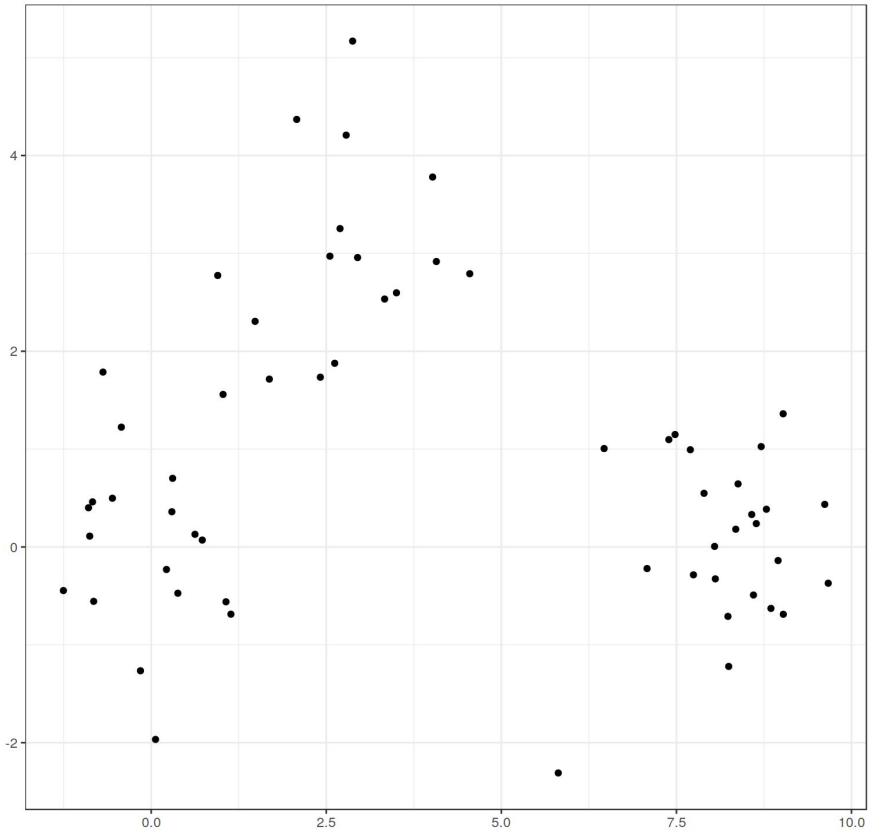


Recommender Systems

Many recommender systems use clustering to filter their user base (you) into clusters. What other TV shows are other people in your cluster watching?

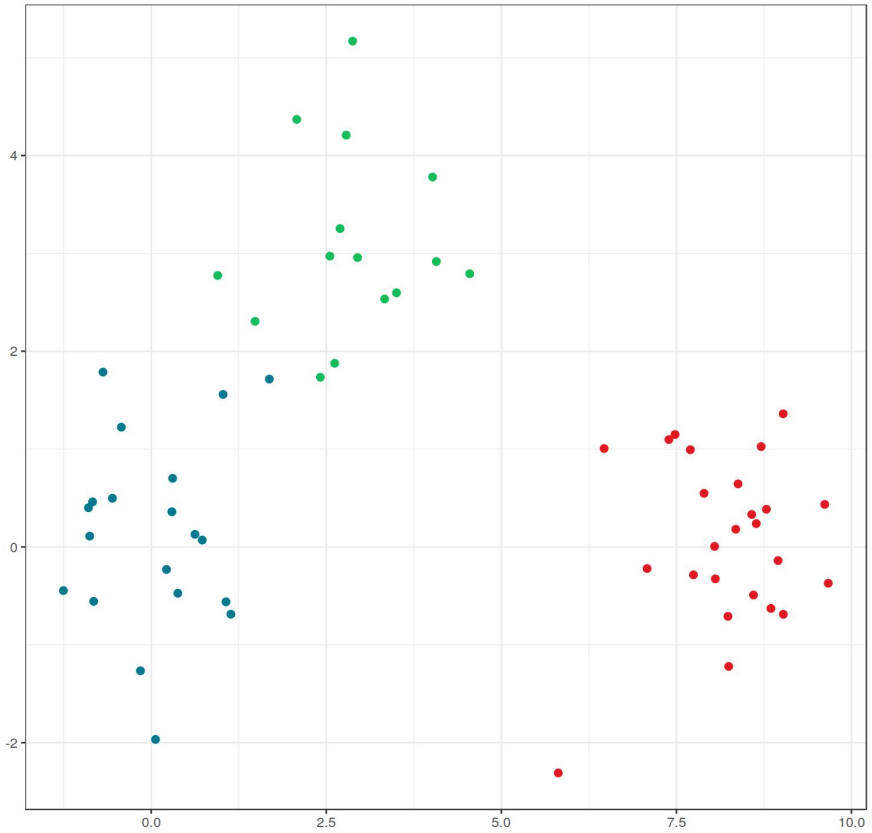


Geometrically, what are we doing?



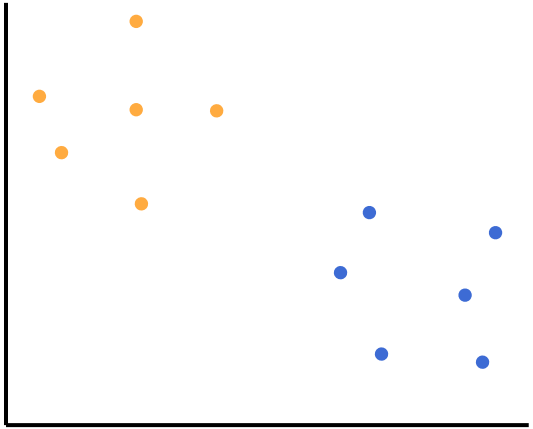
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Geometrically, what are we doing?

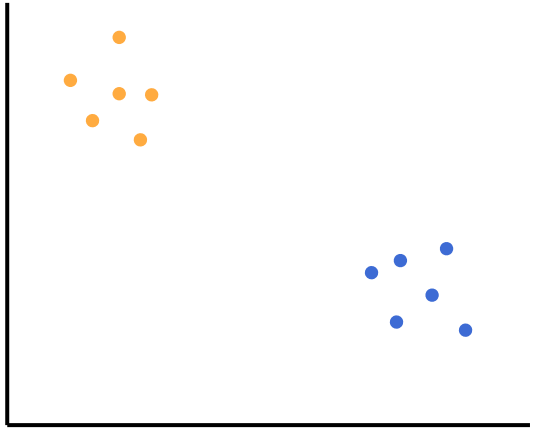


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Which of these clusterings is “better?” Why? How might we measure that?



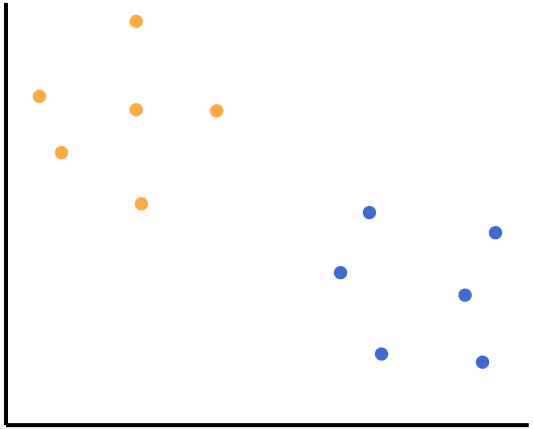
A



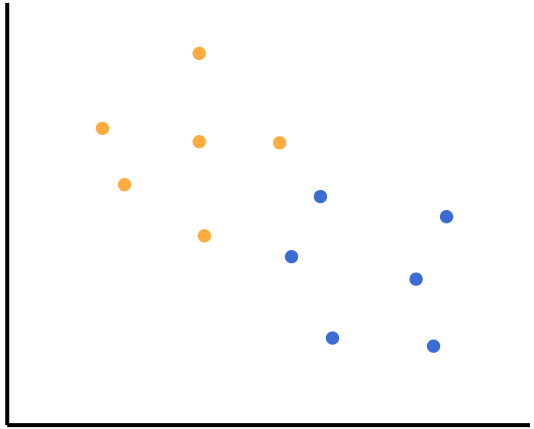
B

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Which of these clusterings is “better?” Why? How might we measure that?



A



B

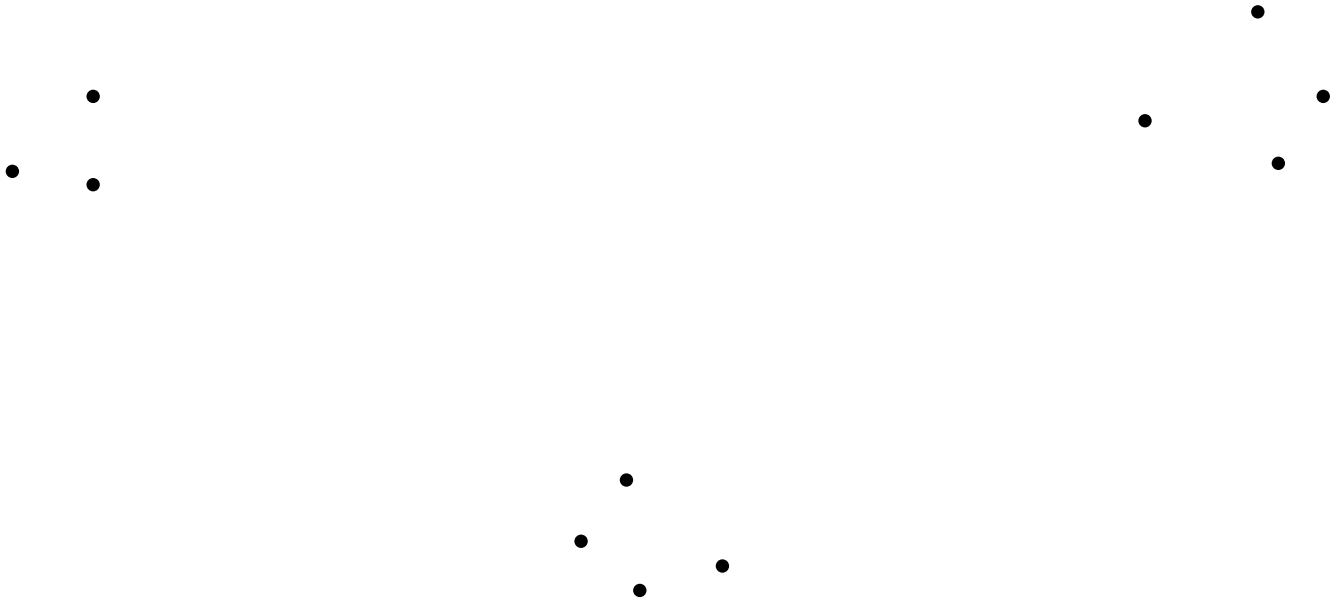
k-Means Clustering

There are many clustering algorithms. We're going to focus on ***k*-means clustering**.

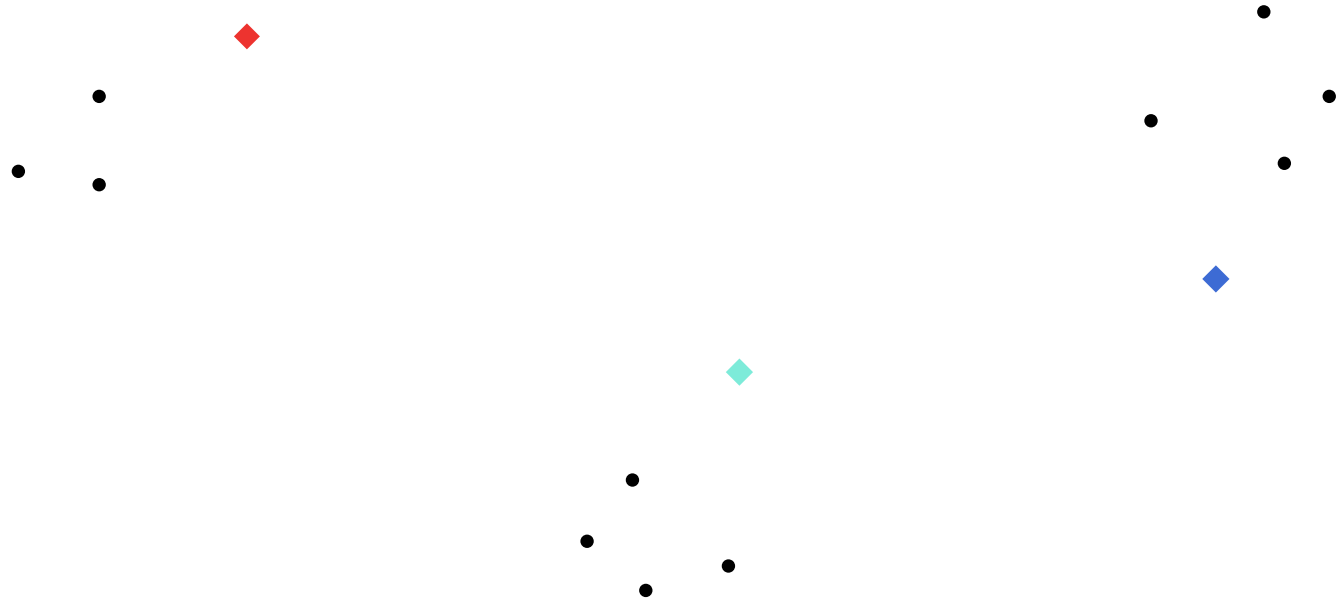
k-means clustering is the most popular and easiest to understand. However, it does have a few shortcomings.

To start, the only thing we need is to decide on *k*, the number of clusters into which we'd like to group our data.

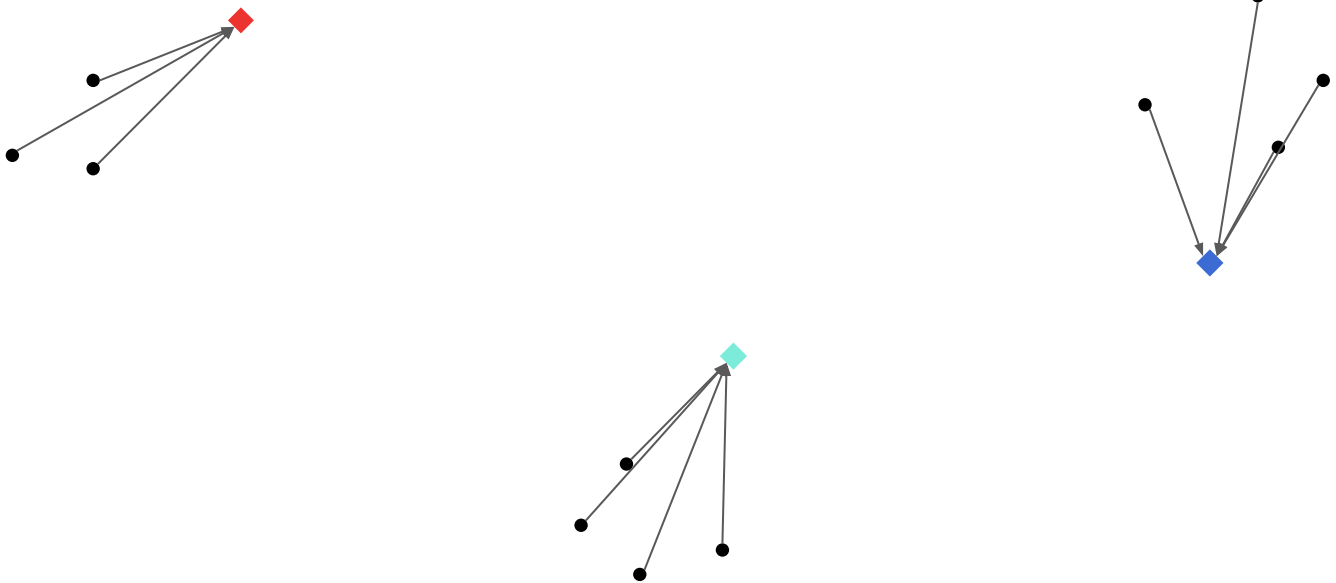
Step 0: Here's our data! Pick *k*.



Step 1: Randomly initialize *k* “centroids.”



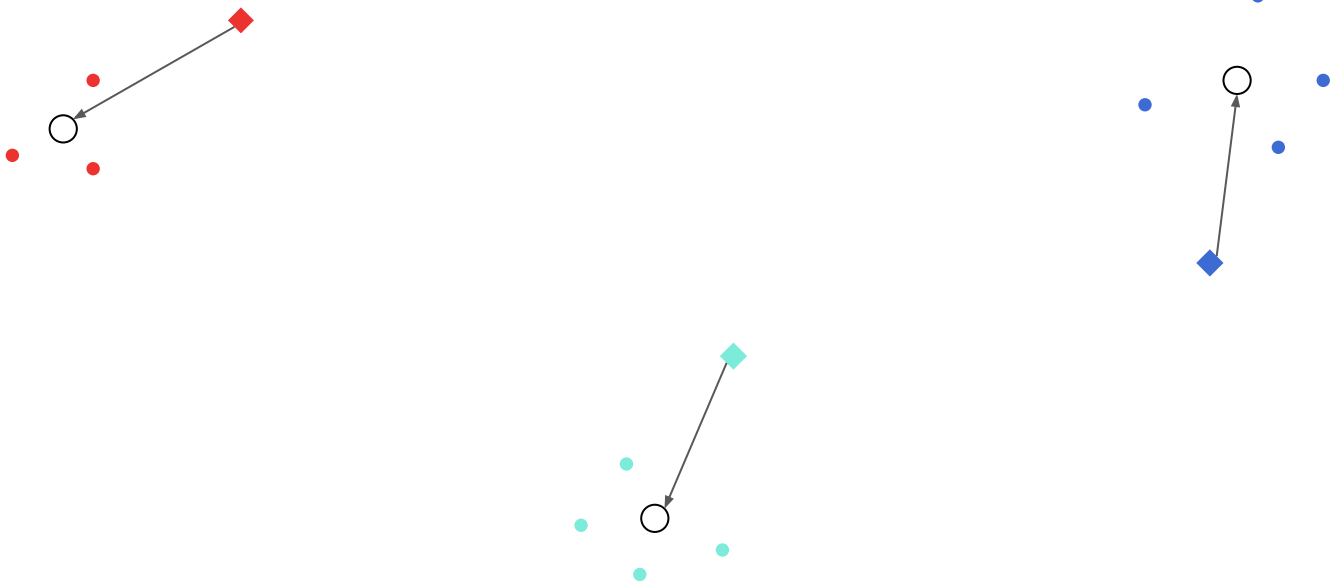
Step 2: Each observation is assigned to be in the cluster associated with the nearest centroid.



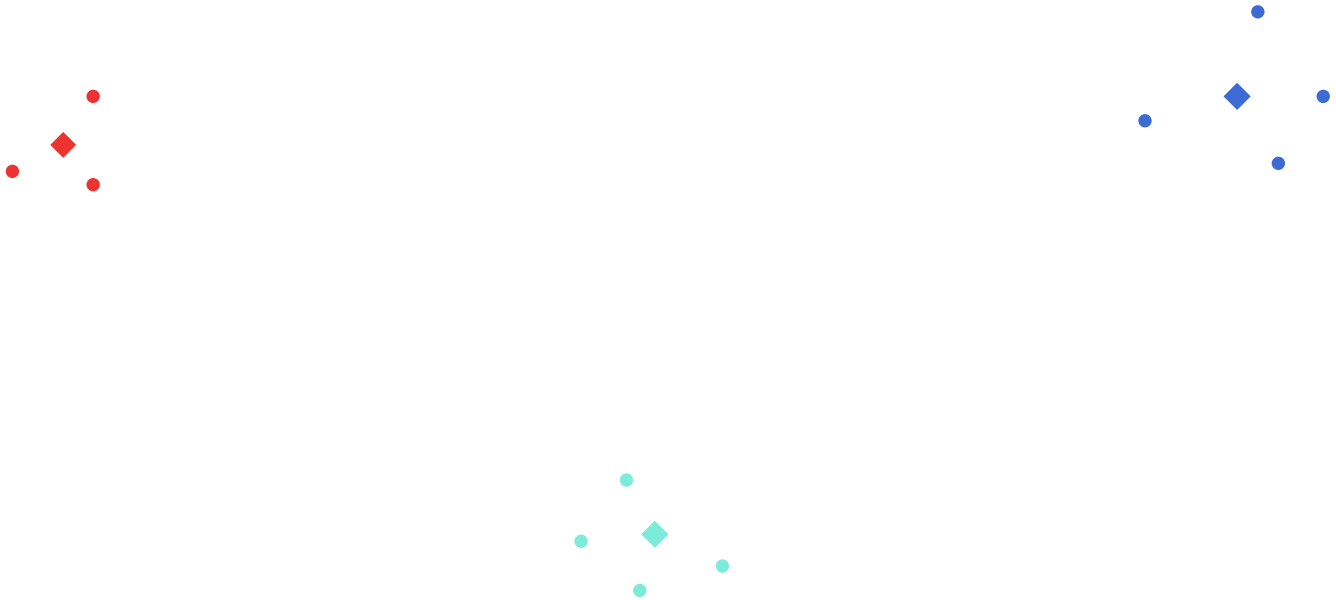
Step 2: Each observation is assigned to be in the cluster associated with the nearest centroid.



Step 3: Each centroid is moved to its cluster's center of mass.



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Step 4: Repeat steps 2 through 3 until convergence!





How much does _____ matter in *k*-Means? Applet!

k

How our data are shaped

Initial centroid placement

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When is *k*-means good?

k-means is best when:

- Your clusters are blob-shaped (as opposed to U-shaped)
- Your clusters are more circular (as opposed to a stripe shape)
- Your blob-shaped, circular clusters are approximately the same size

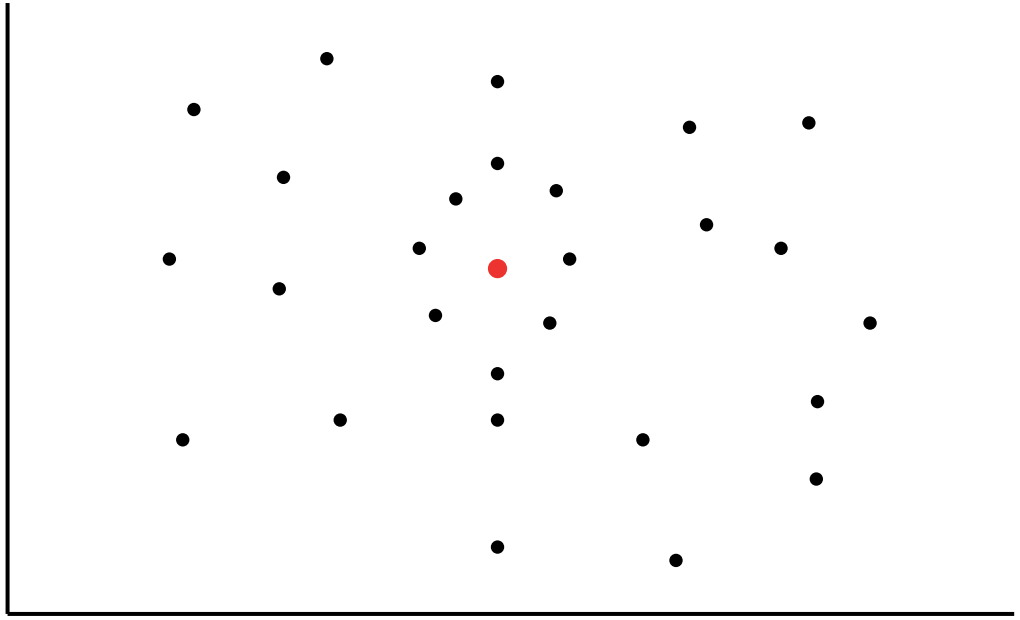
These conditions are pretty much never met with real data. Nonetheless, *k*-means usually performs really well on imperfect data!

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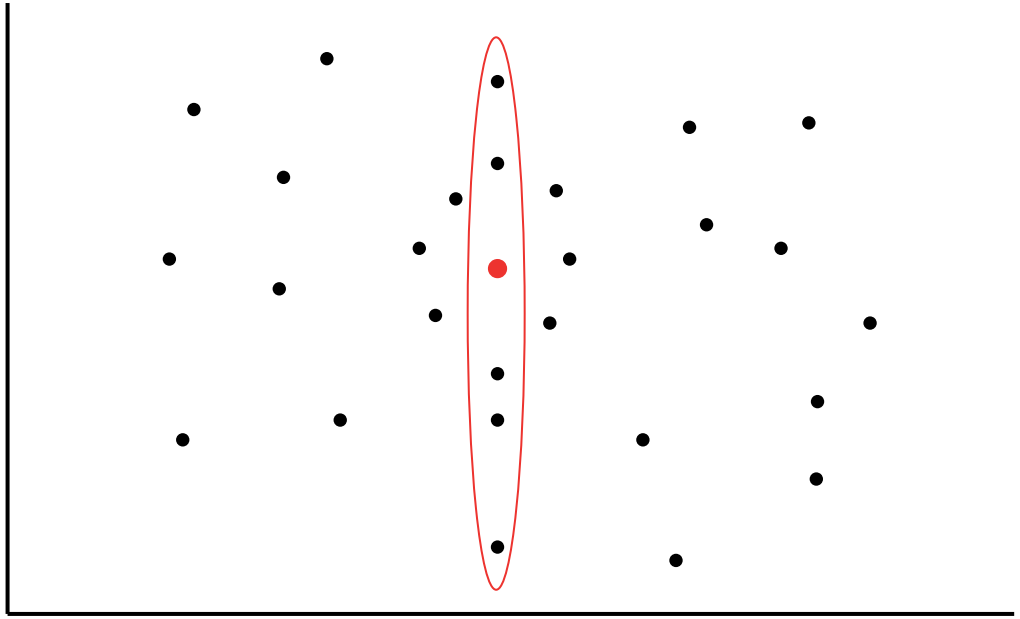
One last thing before coding...



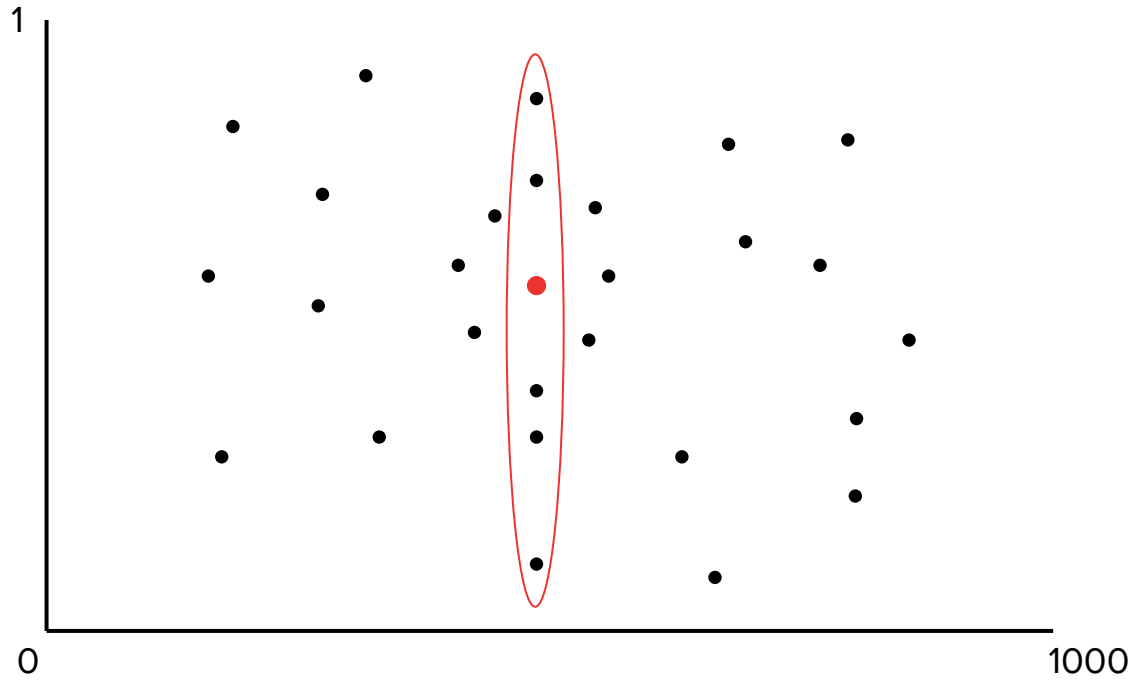
What are the red point's 5 nearest neighbors?



Actually, it's these! How?



The scale of the *x*-axis is way larger! We need to rescale!



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Let's try it out!



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